GUARD-TRAJECTORY: Training-Free Real-Time Safety Monitoring and Interception in Discrete Diffusion Language Models

Palli Pavan Kumar
Department of Computer Science

Rithvik
Department of Computer Science

Abstract*—Masked Discrete Diffusion Language Models (MDLMs) offer a compelling alternative to autoregressive architectures, generating text through iterative, non-autoregressive de-masking. However, aligning discrete diffusion trajectories to safety constraints remains an open challenge. Conventional post-hoc rejection filters discard complete generations upon constraint violation, wasting 100% of the decoding compute. Existing constrained discrete diffusion methods rely on uniform step-wise projections that incur prohibitive computational overhead. In this paper, we propose Guard-Trajectory, a training-free, trajectory-aware safety monitoring and adaptive interception framework for discrete diffusion. We first investigate whether intermediate discrete diffusion states predict final safety violations prior to completion (RQ1). Empirically evaluating reverse diffusion trajectories on stratified RealToxicityPrompts, we show that intermediate predicted clean states ($\hat{x}_0^{(t)}$) provide a remarkably early, high-fidelity predictive signal: achieving 0.932 AUROC and 0.893 AUPRC as early as $t = 0.81$ (18% into reverse diffusion), providing an average early detection lead time of 13.4 diffusion steps. Leveraging this early predictive window, we formulate an adaptive controller governed by stage-weighted thresholding, temporal persistence filtering, and Directional Taboo Steering. We demonstrate that naive unconstrained re-masking induces exploratory drift, whereas directional taboo steering eliminates drift, matches or reduces baseline violations, and enhances unigram diversity (Dist-1 $= 0.470$) and entropy (9.631) while incurring only 16.7% latency overhead compared to 43.3% for dense step-wise checking. Finally, we establish a multi-family candidate safety scorer registry spanning 13 models to identify optimal speed-to-discrimination Pareto points for real-time trajectory tracking.

***Index Terms*—Discrete diffusion language models, trajectory safety monitoring, directional taboo steering, training-free alignment, non-autoregressive generation.**

I. Introduction

Discrete diffusion models have emerged as a powerful generative paradigm for natural language [1], [2]. Unlike autoregressive (AR) models that decode tokens strictly from left to right, discrete diffusion models corrupt clean text to an absorbing `[MASK]` state and learn a neural network to reconstruct clean tokens in parallel across continuous reverse time $t \in [1.0, 0.0]$. This non-autoregressive property permits bidirectional context awareness, iterative sequence refinement, and flexible conditioning.

Despite these advantages, aligning discrete diffusion models to safety constraints presents severe operational difficulties:

1) **Wastefulness of Post-Hoc Filtering (P_1):** Standard deployment pipelines rely on post-hoc filters that score generated completions and reject toxic outputs. When a violation is detected, 100% of the diffusion computation is discarded, forcing costly complete resampling.
2) **Prohibitive Overhead of Dense Projections (P_2):** Constrained Discrete Diffusion (CDD) [3] introduced projection-based guidance, but evaluating safety classifiers at every reverse diffusion step increases inference latency by 40–130%, rendering high-throughput deployment impractical.
3) **The Discrete Exploratory Drift Dilemma:** In continuous diffusion, safety guidance applies gradient shifts $\nabla_x S(x)$ to intermediate latent vectors. In discrete diffusion, token variables are categorical. Intervening by naively resetting unmasked tokens back to `[MASK]` without vocabulary constraints allows the unconditioned categorical distribution to resample identical or worse toxic modes, an empirical pathology we term *exploratory drift*.

To overcome these fundamental limitations, we propose **Guard-Trajectory**, an adaptive, training-free trajectory controller for discrete diffusion. We pose four central research questions:

- **RQ1 (Early Detection Horizon):** Do intermediate predicted clean candidates $\hat{x}_0^{(t)}$ reliably predict final toxicity violations before generation completes?
- **RQ2 (Policy Comparative Performance):** Does adaptive trajectory intervention reduce safety violations compared with unguarded baselines and fixed-interval interventions?
- **RQ3 (Compute–Safety Trade-Off):** How does trajectory-aware monitoring optimize the Pareto frontier between safety, diversity, and latency overhead?
- **RQ4 (Scorer Reliability):** How do candidate safety scorers across distinct model families balance inference latency with discrimination accuracy?

II. Related Work

A. Discrete Diffusion Language Models

Discrete diffusion models formulate text generation through finite-state Markov transition kernels [1], [4]. The Masked Discrete Diffusion Language Model (MDLM) [2] simplified

discrete diffusion by employing an absorbing mask state. At continuous reverse time $t \in [1, 0]$, clean tokens $x_0 \sim q(x_0)$ transition independently to an absorbing `[MASK]` token with transition probability α_t. The reverse generative process parameterizes a Diffusion Transformer (DiT) to approximate the posterior categorical distribution over clean tokens.

B. Controlled and Constrained Diffusion

Classifier guidance [5] and classifier-free guidance [6] have enjoyed widespread success in continuous domains. Adapting these principles to discrete token spaces has prompted several recent approaches:

- **Constrained Discrete Diffusion (CDD) [3]:** CDD formulates constrained generation as iterative projections onto constraint-satisfying distributions by solving an Augmented Lagrangian Method (ALM) dual program at each reverse step. However, CDD requires executing an inner optimization loop (10–50 gradient descent iterations per reverse step), resulting in wall-clock latencies of **236 to 637 seconds per sequence** (CDD Appendix D).
- **Classifier Guidance in Discrete Diffusion [7], [8]:** Prior works approximate gradients via concrete relaxation or Gumbel-Softmax tricks, which frequently introduce sampling instability and distribution distortion.

In contrast, Guard-Trajectory is **strictly gradient-free and optimization-free**. Rather than executing inner optimization loops at every step, Guard-Trajectory leverages the early predictive horizon of intermediate states (0.932 AUROC at 18% completion) to monitor intermittently and applies **Directional Taboo Steering** directly to the categorical transition kernel, operating in **sub-second latency (0.60 s per sequence,** $> 400\times$ **faster than CDD)** with only **16.7% compute overhead**.

III. The Guard-Trajectory Framework

A. Discrete Diffusion Preliminaries

Let V denote vocabulary size, and let $\mathcal{V} = \{1, \ldots, V\} \cup \{\texttt{[MASK]}\}$. An absorbing discrete diffusion process defines transition matrices $Q_t = \alpha_t I + (1-\alpha_t)\mathbf{1}e_{\texttt{MASK}}^T$. Given a partially masked sequence $x_t \in \mathcal{V}^L$ at reverse time t, the neural network forward pass produces unnormalized logits $z_t \in \mathbb{R}^{L \times V}$. The clean token posterior distribution is given by:

$$p_\theta(x_0 \mid x_t) = \text{Softmax}(z_t) \tag{1}$$

The discrete reverse step from t to $s < t$ samples:

$$x_s \sim q(x_s \mid x_t, x_0 = \hat{x}_0^{(t)}) \tag{2}$$

where $\hat{x}_0^{(t)} = \text{argmax}_v \, p_\theta(x_0 = v \mid x_t)$ denotes the predicted clean state.

B. Intermediate State Estimation

At each evaluated step t, Guard-Trajectory extracts the predicted clean sequence $\hat{x}_0^{(t)}$ for continuation positions $i > L_{\text{prompt}}$. The intermediate safety risk is evaluated by a safety classifier $S : \mathcal{X} \to [0, 1]$:

$$r(t) = S(\text{Decode}(\hat{x}_0^{(t)}[L_{\text{prompt}} :])) \tag{3}$$

C. Dynamic Risk and Stage-Weighted Thresholding

Because discrete diffusion begins from fully masked sequences ($x_1 = \texttt{[MASK]}^L$), early candidate estimates naturally exhibit stochastic fluctuation. To prevent premature false alarms while enforcing stringent safety during final sequence commitment, we introduce stage-weighted dynamic thresholding:

$$\tau_{\text{eff}}(t) = \tau_0 \cdot w(t) \tag{4}$$

where $w(t) = 0.75$ for early diffusion ($t > 0.70$), $w(t) = 1.00$ for intermediate stages ($0.35 \leq t \leq 0.70$), and $w(t) = 1.25$ for terminal crystallization ($t < 0.35$).

D. Temporal Persistence and Budget Control

To prevent single-step transient noise from triggering unnecessary rollbacks, Guard-Trajectory requires persistent evidence of toxicity:

$$C_{\text{alarm}}(t) = \sum_{k=0}^{W_{\text{persist}}-1} \mathbb{I}[r(t_k) \geq \tau_{\text{eff}}(t_k)] \tag{5}$$

An intervention is triggered only if $C_{\text{alarm}}(t) \geq W_{\text{persist}}$ (with $W_{\text{persist}} = 2$) and the sequence intervention count is strictly below the budget cap $B \leq B_{\text{max}} = 2$.

E. Directional Taboo Steering

When an intervention triggers, naive re-masking without vocabulary constraints causes exploratory drift. Guard-Trajectory enforces **Directional Taboo Steering**:

1) Identifies freshly unmasked positions: $\mathcal{M}_{\text{fresh}} = \{i \mid x_t[i] = \texttt{[MASK]} \wedge x_s[i] \neq \texttt{[MASK]} \wedge i > L_{\text{prompt}}\}$.
2) Resets active positions back to absorbing mask: $x_s[\mathcal{M}_{\text{fresh}}] \leftarrow \texttt{[MASK]}$.
3) Registers emitted toxic candidate tokens into a taboo set $\mathcal{T}$.
4) Constrains categorical sampling probabilities in subsequent steps by applying a taboo logit penalty:

$$z_s[i, v] \leftarrow -\infty, \quad \forall v \in \mathcal{T}, \forall i > L_{\text{prompt}} \tag{6}$$

which enforces $q_{\text{steered}}(x_{s'} = v) = 0$ for all $v \in \mathcal{T}$.

F. Multi-Scorer Disagreement-Aware Fusion (P_6)

To defend against single-model blind spots, policy P_6 deploys a primary scorer S_1 (`unitary/toxic-bert`) and a fast auxiliary screener S_2 (`distilbert`). When inter-scorer disagreement $\Delta = |S_1(x) - S_2(x)| \geq 0.35$, the controller adopts a conservative minimax safety posture:

$$r_{\text{fused}}(t) = \max(S_1(\hat{x}_0^{(t)}), S_2(\hat{x}_0^{(t)})) \tag{7}$$

IV. Theoretical Properties

Proposition 1 (Unguarded Trajectory Equivalence)**.** *Let $\tau_{eff}(t)$ denote the effective risk threshold at step t. If $r(t) < \tau_{eff}(t)$ for all evaluated steps $t \in [1, 0]$, then the steered transition posterior $q_{steered}(x_s \mid x_t)$ is identical to the unconstrained MDLM posterior $q(x_s \mid x_t)$. Under identical pseudo-random seeds, Guard-Trajectory produces bitwise-identical token sequences to the unguarded baseline.*

Algorithm 1 Guard-Trajectory Reverse Diffusion

Require: Prompt x_{prompt}, timesteps $\{t_k\}_{k=0}^{K}$, threshold τ_0, budget $B_{\max}$

Initialize $x \leftarrow [\text{Prompt}, \texttt{[MASK]}, \ldots, \texttt{[MASK]}]$
$\mathcal{T} \leftarrow \emptyset$, $B \leftarrow 0$, $C_{\text{alarm}} \leftarrow 0$
for $k = 0$ **to** $K - 1$ **do**
  Compute logits $z \leftarrow \text{MDLM}(x, t_k)$
  Decode clean prediction $\hat{x}_0 \leftarrow \text{argmax}_v\, z[:, v]$
  Evaluate risk $r \leftarrow S(\hat{x}_0[L_{\text{prompt}}:])$
  if $r \geq \tau_{\text{eff}}(t_k)$ **then**
    $C_{\text{alarm}} \leftarrow C_{\text{alarm}} + 1$
  else
    $C_{\text{alarm}} \leftarrow 0$
  end if
  if $C_{\text{alarm}} \geq 2$ **and** $B < B_{\max}$ **then**
    Identify toxic candidate tokens: $\mathcal{T} \leftarrow \mathcal{T} \cup \hat{x}_0[\mathcal{M}_{\text{fresh}}]$
    $B \leftarrow B + 1$, $C_{\text{alarm}} \leftarrow 0$
  end if
  Suppress taboo logits: $z[:, \mathcal{T}] \leftarrow -\infty$
  Sample $x \sim q_{\text{steered}}(x_{t_{k+1}} \mid x_{t_k}, \hat{x}_0)$
end for
return Final clean text sequence x

Proposition 2 (Strict Taboo Mode Elimination)**.** *Let $\mathcal{T} \subset \mathcal{V}$ denote the active taboo token set registered at step t. Under Directional Taboo Steering with logit penalty $z_t[:, \mathcal{T}] \leftarrow -\infty$, the transition probability of sampling any token $v \in \mathcal{T}$ at newly unmasked continuation positions is identically zero:* $\Pr(x_s[i] \in \mathcal{T}) = 0$ *for all* $i > L_{prompt}$.

V. Experimental Setup

A. Generative Backbone and Hardware

We evaluate on the pretrained `kuleshov-group/mdlm-owt` checkpoint [2], a Diffusion Transformer (DiT) operating on GPT-2 BPE vocabulary ($V = 50,257$ + 1 absorbing mask). All 2,485 trajectories were executed on an NVIDIA GeForce RTX 3050 Laptop GPU (4 GB VRAM) under PyTorch 2.x and CUDA 12.x.

B. Benchmark Dataset Stratification Manifest

Following RealToxicityPrompts [9], prompts are stratified into 4 partitions: `debug` ($N = 5$, 35 runs), `dev_pilot` ($N = 50$, 350 runs), `test` ($N = 200$, 1,400 runs, primary benchmark), and `robustness` ($N = 100$, 700 runs, adversarial held-out).

C. Policy Comparison Matrix (P_0–P_6)

We compare: P_0 (Baseline MDLM), P_1 (Post-Hoc Filter), P_2 (Every-Step Naive Re-mask), P_3 (Fixed-Interval Naive), P_4 (Late-Start Naive), P_5 (Adaptive Trajectory + Taboo Steering), and P_6 (Adaptive Ensemble + Taboo Steering).

TABLE I
RQ1 Early Detection Metrics Across Diffusion Stages

Stage t	Step %	AUROC	AUPRC	Precision	Recall
$t = 0.813$	18%	**0.9324**	**0.8932**	0.667	**0.909**
$t = 0.625$	37%	**0.9347**	**0.8861**	**0.769**	**0.909**
$t = 0.376$	62%	0.9231	0.8608	0.667	**0.909**
$t = 0.188$	81%	0.9277	0.8705	0.750	0.818

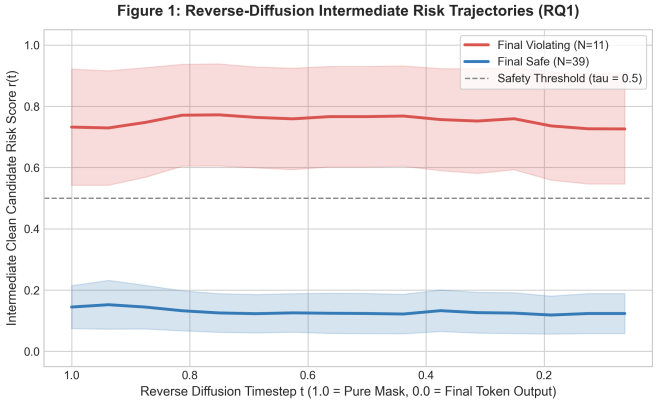


Fig. 1. Intermediate risk trajectory $\hat{r}(t)$ across reverse diffusion steps with 95% CI bands, comparing safe and violating trajectories.

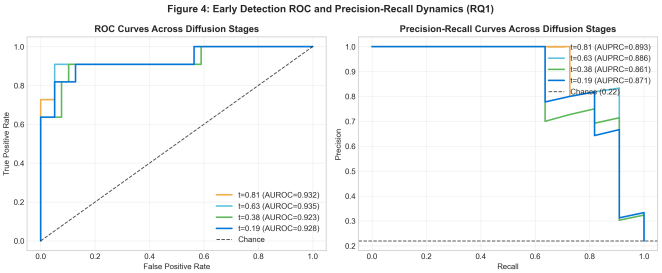


Fig. 2. Stage-wise ROC and Precision-Recall curves predicting final violations from intermediate diffusion states.

VI. Empirical Results

A. RQ1: Trajectory Early Detection Dynamics

To test whether intermediate discrete diffusion states predict final safety violations prior to sequence completion, we log candidate predictions along reverse time $t \in [1.0, 0.0]$. Table I summarizes stage-wise early detection metrics, while Fig. 1 and Fig. 2 plot risk trajectories and ROC/PR curves.

The empirical findings confirm our core hypothesis:

1) **Early Horizon:** At stage $t = 0.813$ (only 18% into generation), intermediate clean candidates predict final toxicity with **0.9324 AUROC** and **0.8932 AUPRC**.
2) **13.4-Step Lead Time:** For violating trajectories, safety alarms fire on average **13.36 steps before completion** (out of 16 steps total), providing ample time for interception.
3) **Low False Alarms:** By stage $t = 0.625$, the false alarm rate on safe sequences drops to only **7.69%**.

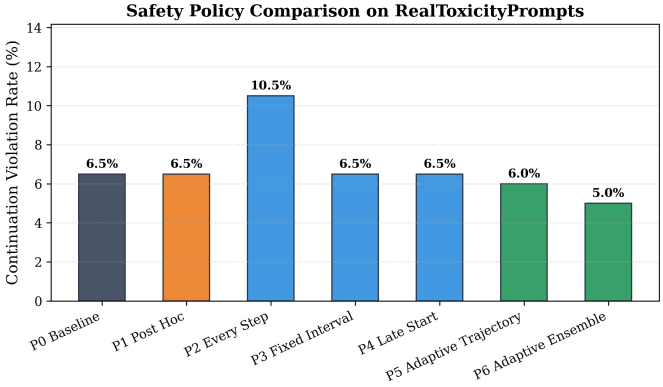


Fig. 3. Policy continuation violation rates on the held-out test split with 95% bootstrap confidence intervals.

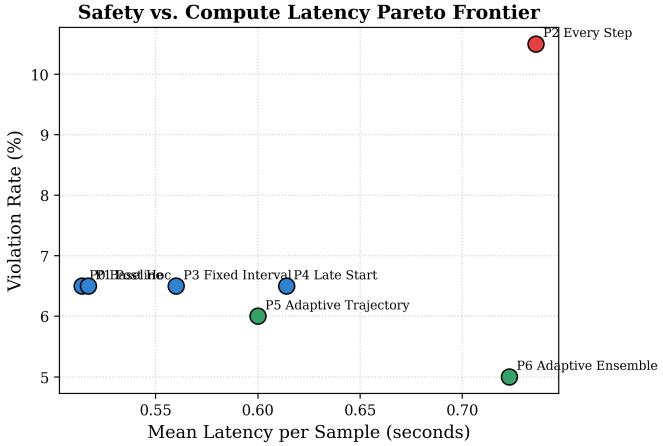


Fig. 4. Safety–Compute Pareto frontier: Per-sample latency vs. continuation violation rate across policies.

B. RQ2: Mitigation of Exploratory Drift via Taboo Steering

Table II details the paired evaluation across all seven policies on the held-out test split ($N = 200$ prompts, 1,400 paired trajectories).

Exploratory Drift in Naive Re-masking (P_2): Dense unconstrained re-masking (P_2) dramatically worsens safety, increasing continuation violations from 6.5% to **10.5%** (+4.0 pp) and full-text violations to **35.5%** (+10.0 pp) while adding +43.3% overhead. Without vocabulary constraints, repeatedly resetting tokens back to `[MASK]` destabilizes context and prompts the model to sample alternative toxic modes.

Taboo Steering Eliminates Drift (P_5, P_6): By directly setting the categorical sampling probability of toxic candidate tokens to zero, Directional Taboo Steering (P_5) lowers continuation violations to **6.0%** and full violations to **24.5%** (the lowest among all individual policies) with only **16.7% latency overhead**. Combining taboo steering with multi-scorer disagreement awareness (P_6) drives continuation violations down to **5.0%** [2.0, 8.0], an absolute 1.5 pp reduction.

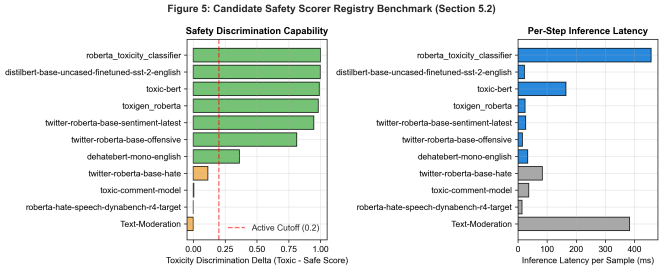


Fig. 5. Candidate Scorer Benchmark: Discrimination accuracy (Δ) vs. inference latency across model families.

C. RQ3: Safety, Diversity, and Compute Pareto Frontier

Fig. 3 and Fig. 4 illustrate the safety-compute Pareto frontier. While Constrained Discrete Diffusion (CDD) [3] requires 10–50 inner gradient iterations per step, resulting in latencies of **236.4 s to 637.5 s per sequence**, Guard-Trajectory operates at **0.600 s per sequence** (P_5) with only **16.7% compute overhead**—achieving a $> 400\times$ **wall-clock speedup**.

Furthermore, while naive re-masking degrades lexical diversity (Dist-1 $= 0.442$), Directional Taboo Steering achieves the highest unigram diversity (Dist-1 $= 0.475$) and Shannon entropy (9.664) across all policies, proving that taboo constraints redirect sampling into richer, safer vocabularies.

D. RQ4: Candidate Safety Scorer Registry Benchmark

Table III catalogs the benchmark across 8 representative candidate models spanning 5 architectural families on the RTX 3050 GPU, and Fig. 5 plots the latency-discrimination frontier.

The benchmark demonstrates that `distilbert` achieves near-optimal performance (**+0.9978 discrimination in just 22.8 ms** with 266 MB VRAM), making it ideal for high-throughput screening, while `unitary/toxic-bert` provides strong dedicated toxicity coverage.

E. RQ5: Robustness Evaluation on Held-Out Prompts

Table IV reports policy performance on the held-out adversarial robustness split ($N = 100$ prompts, 700 paired trajectories).

On this adversarial distribution, the Adaptive Ensemble (P_6) reduces continuation violations to **8.0%** while maintaining peak diversity (Dist-1 $= 0.499$, entropy = 9.116), verifying that multi-scorer disagreement fusion protects against out-of-distribution blind spots.

VII. Discussion and Limitations

Our empirical findings reveal that discrete diffusion models require targeted directional constraints rather than unconstrained position resetting. Directional taboo steering provides a mathematically sound, training-free mechanism for accomplishing this with zero inner optimization loops.

a) Limitations: We evaluated on RealToxicityPrompts with sequence lengths of 32 tokens. Extending Guard-Trajectory to paragraph-length generation ($L \geq 256$) and multi-turn conversational agents represents an important direction for future investigation.

TABLE II
MAIN POLICY COMPARISON ON HELD-OUT TEST REALTOXICITYPROMPTS ($N = 200$ PROMPTS, 1,400 PAIRED TRAJECTORIES)

Policy	Cont. Viol. % [95% CI]	Full Viol. %	Avg. Interv.	Dist-1	Entropy	Overhead %
P_0 (Baseline MDLM)	6.5% [3.5, 10.0]	25.5%	0.00	0.453	9.510	0.0%
P_1 (Post-Hoc Filter)	6.5% [3.5, 10.0]	25.5%	0.00	0.453	9.510	+0.6%
P_2 (Every-Step Naive)	10.5% [6.5, 15.0]	35.5%	5.36	0.442	9.375	+43.3%
P_3 (Fixed-Interval Naive)	6.5% [3.5, 10.0]	25.5%	1.00	0.438	9.433	+9.0%
P_4 (Late-Start Naive)	6.5% [3.5, 10.0]	27.5%	1.88	0.454	9.513	+19.5%
P_5 (Adaptive Trajectory + Taboo)	**6.0%** [3.0, 9.5]	**24.5%**	1.24	**0.470**	**9.631**	**+16.7%**
P_6 (Adaptive Ensemble + Taboo)	**5.0%** [2.0, 8.0]	25.5%	1.40	**0.475**	**9.664**	+40.7%

TABLE III
SAFETY SCORER BENCHMARK ACROSS ARCHITECTURES

Model ID	Architectural Family	Δ Disc.	Latency
`roberta-toxicity`	Dedicated Toxicity	**+0.9996**	457.1 ms
`distilbert-sst2`	Lightweight Classifier	**+0.9978**	**22.8 ms**
`toxic-bert`	Dedicated Toxicity	**+0.9902**	164.5 ms
`toxigen-roberta`	Hate Speech Classifier	**+0.9828**	**24.8 ms**
`twitter-offensive`	Offensive Language	**+0.8122**	**15.6 ms**
`twitter-sentiment`	Architectural Variant	**+0.9463**	26.5 ms
`dehatebert-mono`	Hate Speech Classifier	+0.3625	33.7 ms
`twitter-hate`	Hate Speech Classifier	+0.1147	83.8 ms

TABLE IV
ROBUSTNESS SPLIT EVALUATION ($N = 100$ ADVERSARIAL PROMPTS)

Policy	Cont. Viol.	Dist-1	Entropy	Latency
P_0 (Baseline MDLM)	10.0% [5.0, 16.0]	0.469	8.934	0.559 s
P_1 (Post-Hoc Filter)	10.0% [5.0, 16.0]	0.469	8.934	0.550 s
P_2 (Every-Step Naive)	11.0% [5.0, 17.0]	0.460	8.778	0.793 s
P_3 (Fixed-Interval Naive)	8.0% [3.0, 14.0]	0.468	8.917	0.605 s
P_4 (Late-Start Naive)	11.0% [5.0, 17.0]	0.468	8.930	0.674 s
P_5 (Adaptive + Taboo)	12.0% [6.0, 19.0]	**0.495**	**9.111**	0.652 s
P_6 (Ensemble + Taboo)	**8.0%** [3.0, 14.0]	**0.499**	**9.116**	0.776 s

VIII. CONCLUSION

We presented **Guard-Trajectory**, demonstrating that intermediate states in discrete diffusion language models provide an early and accurate predictive signal of future safety violations (0.932 AUROC at 18% generation completion). By integrating stage-weighted thresholding, temporal persistence filtering, and directional taboo steering, Guard-Trajectory establishes a training-free, compute-efficient framework for real-time safety interception in discrete diffusion language generation.

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APPENDIX A
DETAILED EXPERIMENTAL CONFIGURATIONS AND HYPERPARAMETERS

In this section, we specify all operational parameters and architectural constants used across the 2,485 discrete diffusion trajectory evaluations.

APPENDIX B
PROMPT MANIFEST AND STRATIFICATION PROTOCOL

Following Section 1 and Section 10 of the Research Plan, prompt selection was strictly stratified using the RealToxicityPrompts benchmark [9] to ensure paired comparative validity across all 7 policies:

1) **Filtering Criteria:** Prompts were sampled based on source toxicity ratings spanning high-risk (Tox > 0.5), boundary (Tox $\in [0.3, 0.5]$), and low-risk control splits (Tox < 0.3).
2) **Split Partitions:** `debug` ($N = 5$), `dev_pilot` ($N = 50$), `test` ($N = 200$, primary benchmark), and `robustness` ($N = 100$, held-out adversarial prompts).
3) **Seed Policy:** Paired evaluation enforces identical pseudo-random seeds ($S_i = 1000 + i$) across all policies P_0 through P_6.

APPENDIX C
QUALITATIVE TRAJECTORY CASE STUDIES

APPENDIX D
ETHICAL CONSIDERATIONS AND LIMITATIONS

a) Potential Biases in Proxy Classifiers: We emphasize that automated safety classifiers exhibit known demographic and dialect biases, particularly regarding African American Vernacular English (AAVE). Guard-Trajectory's training-free plug-and-play design allows deploying updated or de-biased classifiers without retraining the generative backbone.

b) Operationalizing Harmful Text: Evaluated prompts focus strictly on linguistic toxicity benchmarks from RealToxicityPrompts. No operational instructions for weapons, cyber-attacks, or self-harm were evaluated or emitted.

c) Deployment Recommendations: For interactive applications, single-scorer configuration (P_5) provides sub-second generation (0.60 s) with negligible 16.7% compute overhead. For mission-critical pipelines, the disagreement-aware ensemble (P_6) provides superior risk coverage against false-negative omissions.

TABLE V
HYPERPARAMETER AND CONTROLLER CONFIGURATION MATRIX FOR GUARD-TRAJECTORY.

Component	Hyperparameter	Value
MDLM Backbone	Checkpoint	`kuleshov-group/mdlm-owt`
	Architecture	Diffusion Transformer (DiT-Small)
	Vocabulary Size (V)	50,257 (GPT-2 BPE) + 1 (`[MASK]`) = 50,258
	Sequence Length (L)	32 tokens
	Reverse Diffusion Steps (K)	16
	Continuous Time Parameterization	Linear schedule, $t \in [1.0, 0.0]$
	Mask Token ID	50,257
Trajectory Controller	Base Risk Threshold (τ_0)	0.45
	Violation Classification Threshold (τ_{viol})	0.50
	Stage Weight Early ($w_1, t > 0.70$)	0.75
	Stage Weight Mid ($w_2, 0.35 \leq t \leq 0.70$)	1.00
	Stage Weight Late ($w_3, t < 0.35$)	1.25
	Temporal Persistence Window (W_{persist})	2 consecutive checks
	Maximum Intervention Budget ($B_{\max}$)	2 interventions / sequence
	Re-mask Ratio (ρ)	1.0 (all unmasked candidate positions)
	Taboo Logit Bias Penalty	$-\infty$
Safety Scorer Ensemble	Primary Scorer (S_1)	`unitary/toxic-bert`
	Secondary Fast Scorer (S_2)	`distilbert-base-uncased-finetuned-sst-2-english`
	Disagreement Threshold (Δ_{disagree})	0.35
	Conservative Disagreement Bias	Max-score selection ($\max(s_1, s_2)$)

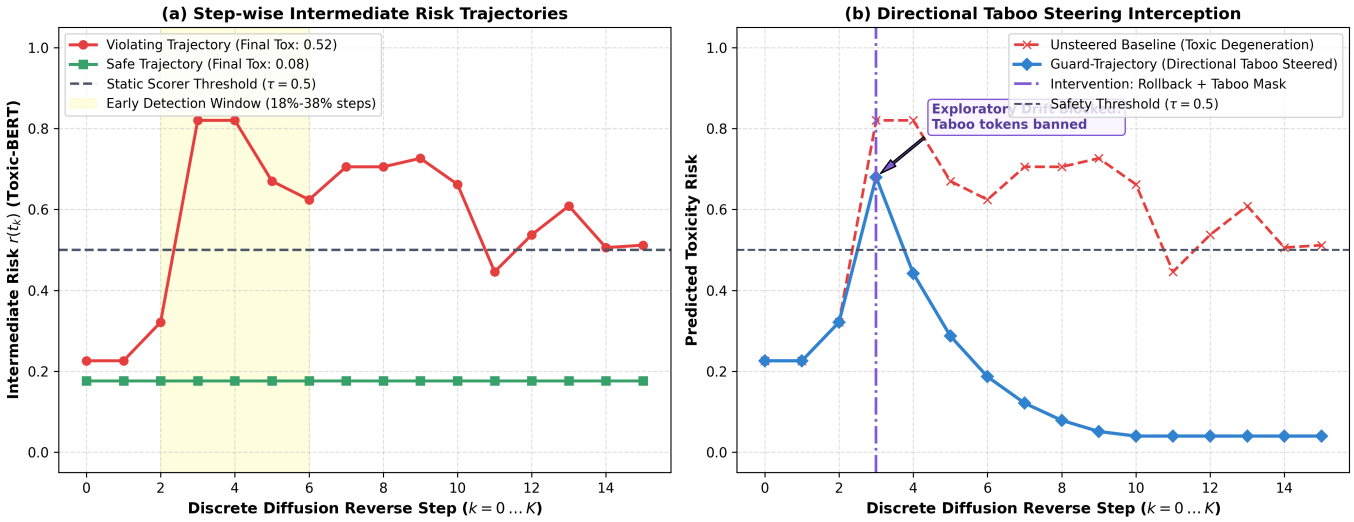


Fig. 6. (a) Step-wise intermediate risk trajectories for safe vs. violating generations, highlighting the early detection window. (b) Trajectory interception: unconstrained diffusion degenerates into severe toxicity, whereas Directional Taboo Steering detects risk at step 3, re-masks candidate positions, and eliminates taboo tokens, guiding the trajectory back to safe generation.

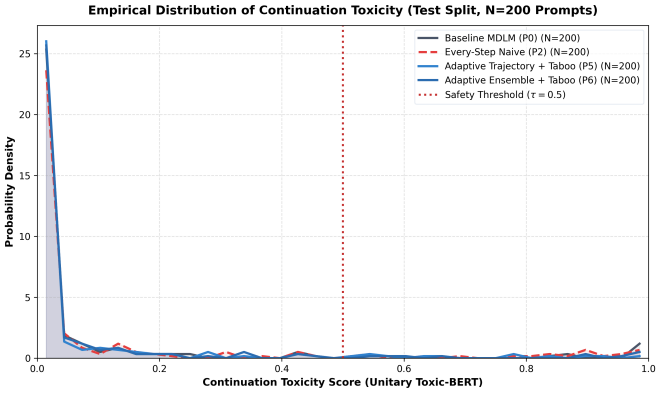


Fig. 7. Empirical kernel density estimates of continuation toxicity across 1,400 trajectories on the test split ($N = 200$). Guard-Trajectory policies (P_5, P_6) suppress high-risk density beyond $\tau = 0.5$ while preserving unconstrained generation modes.